

Brain Rhythms in Cognition and Self-Aware Networks

An Atomized Four-Lens Genealogy of Oscillatory Rendering, Traveling Waves, and Brain-Body Integration

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Abstract

Neural oscillations, synchrony, dendritic integration, coherence-based routing, traveling waves, and brain-body coupling belong to a long scientific lineage. Self-Aware Networks (SAN) does not require those established components to have originated with Micah Blumberg. Its narrower priority claim concerns the way they were joined. Publicly dated records from 2011 and 2012 describe a predictive self and internally constructed world without a hidden observer, recursive multimodal brain-computer feedback, temporally aligned neural events, and the mind as a projector of a continuously updated world model. Public 2017 records add spatial computation in neuroscience, a time-varying three-dimensional ionic matrix, coincidence as a neural information unit, dendritic computation, recurrent thalamocortical rendering, excitation and inhibition as co-equal information, and a difference-only transmission operation. The 2022 SAN corpus then names and consolidates Phase Wave Differential (PWD), relative tonic / phasic rendering, and Neural Array Projection Oscillation Tomography (NAPOT).

This paper compares that staged genealogy with *Brain rhythms in cognition - controversies and future directions*, a 35-author position paper submitted in 2025, and with four of its relevant primary-study routes. The comparison uses four lenses: operational mechanism, causal / narrative architecture, conclusion-backward forensic analysis, and longitudinal signature migration. It also provides a category-theoretic and process-calculus translation. The result is split by atom. Earlier researchers retain priority for exact components including oscillatory binding, temporal correlation, large-scale phase synchronization, nested rhythms, coherence routing, dendritic integration, and specific traveling-wave mechanisms. Blumberg retains the earlier dated priority for the bounded joined architecture that moves from an ongoing oscillatory context through receiver-relative change and distributed projection to an internally rendered predictive model, action, and returned sensory evidence. The 2025 position paper later assembles phase, amplitude, cross-frequency interaction, traveling waves, brain-body coupling, perception, attention, memory, communication, and a proposed unified framework, but it does not instantiate the complete PWD or NAPOT operators. That incompleteness caps identity; it does not erase partial operational and narrative equivalence.

No evidence reviewed here establishes access, copying, or misconduct. The target bibliography contains no occurrence of “Blumberg,” “Self-Aware,” or “NAPOT,” which establishes a citation-coverage gap only. A numerical independent-derivation probability is not reported because the required background frequencies and dependency model do not yet exist. The calibrated status is therefore **quantitative-not-calibrated**, while the historical finding is **Micah-earlier on the joined SAN architecture, split priority by atom**.

Scope and evidence boundary

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1. The ordinary problem: a stable world assembled by a moving body

When a person turns their head, reaches for a cup, or follows a moving object, the retinal image, muscle state, eye position, and incoming sensory evidence all change. Experience nevertheless presents a relatively stable scene containing persistent objects, relations, and possible actions. The brain must continually solve at least five problems:

1. distinguish new input from activity already underway;
2. combine signals that arrive at different places and times;
3. preserve spatial relations while the observer moves;
4. select some signals for amplification, inhibition, memory, or action; and
5. compare the resulting action with the sensory evidence that returns.

Established neuroscience supplies many pieces of this account. Oscillations organize excitability; phase relations affect when inputs are effective; coherence can alter communication; dendrites combine contextual and feedforward input; traveling waves move structured activity through space; and bodily rhythms can influence neural tissue. Those facts do not by themselves explain how a distributed biological network becomes the process that both constructs and undergoes its internal model.

SAN proposes a particular answer. A receiving population is never blank. It is already in a state created by its membrane history, synapses, dendrites, inhibition, local rhythms, recurrent input, and bodily context. Incoming activity changes that receiver relative to its existing and expected state. The receiver transforms the change and projects an updated pattern onward. Many such local receive-transform-project events recur across a network. Their joint activity maintains and revises a predictive internal scene, recruits action, and samples the consequences. The network does not send the result to a separate inner viewer: the distributed update process is the observing operation.

The later terms should be earned only after that operation is clear. SAN calls the receiver-relative departure a **Phase Wave Differential** when phase and propagation are part of the measurement. It calls the recurrent distributed projection-and-reconstruction program **Neural Array Projection Oscillation Tomography**. These terms are not synonyms for phase coding or traveling waves. They name additional proposed roles: receiver-relative change as content and distributed recurrent projection as an observer-forming reconstruction process.

2. Comparison question and correction of the prior version

The original 2025 comparison recognized a substantial convergence between SAN and the Keitel et al. position paper, but it made three mistakes. First, it treated several neighboring constructs as if they were identical to PWD or NAPOT. Second, it inferred influence from timing and non-citation without access evidence. Third, it misattributed the 2024 *Science* mechanosensation paper to Kluger and colleagues rather than Luna Jammal Salameh and colleagues.

Draft 4 corrected those errors, but it then overcorrected. It converted named earlier components into author-wide “historical floors” and withdrew the priority analysis for the joined architecture. That is a full-operator-or-zero error: because the early Blumberg stages do not contain every mature PWD / NAPOT detail, and because earlier researchers own component atoms, Draft 4 treated the later joined SAN sequence as if it had no priority-bearing content.

The present paper preserves both sets of corrections:

- PWD is not a synonym for a traveling wave.

- NAPOT is not established by ordinary superposition or coherence.
- non-citation does not prove access, influence, copying, or wrongdoing;
- Jammal Salameh et al. are correctly credited;
- Llinás, Ribary, Singer, Gray, Varela, Lachaux, Rodriguez, Martinerie, Buzsáki, Draguhn, Fries, Larkum, and other prior researchers retain their exact earlier contributions; and
- those exact contributions do not terminate the comparison of a later, differently joined causal architecture.

This audit also consumes Blumberg’s public 26 July 2025 SVGN comparison as a dated first-party discovery router. That article preserves the earlier public comparison argument and several source leads; it is not treated as independent validation of its own conclusion. Every claim used here was re-bounded to the target position paper, the relevant primary studies, and the dated Blumberg sources before entering the atom ledger.

The question is therefore not “Who invented brain rhythms?” It is:

Which exact components were already established, which conjunctions were publicly articulated by Blumberg at each stage, which of those conjunctions appear later in the target program, and what degree of equivalence survives when vocabulary is translated into operations?

3. Method

3.1 Unit of comparison

The unit is a dated proposition or causal atom, not an author, title, topic, keyword, or whole theory. Each row binds:

- an exact or tightly bounded source proposition;
- its date type and provenance grade;
- a neutral operational translation;
- the target proposition and target stage;
- earlier named priors for any component;
- the degree of equivalence;
- the surviving priority delta; and
- a discriminator or falsifier.

An author who published one component earlier does not acquire priority over every later combination containing that component. Conversely, an early functional formulation is not erased because its mature acronym appeared later.

3.2 Full-genealogy saturation

The Micah side was divided into stages rather than represented by the 2022 whitepaper alone:

- 2011 predictive self, whole-network observer, and internally constructed world;
- 2012 temporal alignment, recursive EEG / media feedback, coherent neural sequences, self-world feedback, and projector language;
- 2017 spatial computation, a dynamic 3D/4D internal matrix, coincidence as a bit, dendritic computation, recurrent thalamocortical frames, informational inhibition, and difference-only transmission;
- 2021 recorder evidence for tonic / phasic firing, subthreshold and receiver-state-sensitive perturbation, recurrent population context, cross-modal convergence, closed-loop read / write return, inverse-problem reasoning, interference, and multiscale reconstruction;
- 2022 named NAPOT / PWD, distributed array observation, volumetric rendering, tonic / phasic relations, inhibition-guided selection, and sensorimotor return; and

- 2024-2025 public SAN refinements and the published SAN paper.

Mature terms were not projected backward. The 2011-2017 sources receive credit only for operations actually present in them.

The target side was also staged. A Keitel-led 2022 review discussed rhythms in cognition, phase, power, frequency, rhythmic behavior, and closed-loop experiments. The 2025 position paper later added an explicitly comprehensive roadmap spanning physiological mechanisms, phase, amplitude, phase-amplitude coupling, sampling, traveling waves, brain-body interactions, perception, attention, memory, communication, and a unified framework. This before / after result is attributed to the Keitel-led and overlapping-author program only; it is not automatically imputed to all 35 coauthors.

3.3 Four lenses

Lens 1 - operational mechanism. Do the systems perform the same bounded input-transform-output operation even when their terms differ?

Lens 2 - causal and narrative architecture. Does the same ordered chain of problems, transformations, and consequences recur, even if atoms are renamed, rearranged, or supplemented?

Lens 3 - conclusion-backward forensic analysis. Starting from a later shared conclusion, can the earlier stages be recovered without requiring the later author to use SAN vocabulary? This lens searches for functional equivalence and possible differentiation but does not infer intent.

Lens 4 - longitudinal signature migration. Does a target program's earlier stage lack a joined cluster that appears in its later stage after Blumberg's public formulation? This is a chronology test, not proof of access.

A bounded match survives if any of the first three lenses passes. Lens 4 can strengthen the chronology but cannot alone prove dependence.

3.4 Equivalence grades

This paper uses:

- **exact atom:** the same measurable variable or operation;
- **strong functional equivalence:** the same input, transformation, and output with nonessential implementation differences;
- **partial functional equivalence:** a shared causal subgraph with omitted or changed operators;
- **narrative-architecture equivalence:** the same ordered problem-solution chain without complete mechanistic identity;
- **analogy only:** a useful resemblance that does not preserve the causal operation; and
- **opposite:** incompatible conclusions on the compared proposition.

Different vocabulary or added machinery does not reduce a match by itself. The grade is reduced only when the shared operation, relation, or consequence is genuinely absent.

3.5 Probability boundary

The patent-style probability question is whether the observed joined feature cluster is more expected under dependent than independent development:

$$\Lambda(E) = \log \frac{P(E | D)}{P(E | I)}, \quad (1)$$

where E is the dated evidence cluster, D a dependency hypothesis, and I independent development from the public scientific field.

Equation (1) is a valid comparison target, but a numerical value requires a defined background corpus, feature frequencies, feature dependencies, search protocol, and access model. Those quantities are not available here. Common features such as “oscillations matter” have high background frequency. The joined SAN sequence is less common, but the corpus has not been calibrated well enough to attach a defensible tail probability or “five sigma” label. The probability status is therefore **quantitative-not-calibrated**. This preserves a strong qualitative chronology without manufacturing precision.

4. The established scientific inheritance

The comparison begins by crediting exact prior atoms.

Llinás and Ribary reported coherent 40-Hz thalamocortical activity in human dream states in 1993. Singer and Gray’s 1995 review treated temporal correlation and relative spike timing as candidate mechanisms for visual feature integration. Varela and colleagues made large-scale phase synchronization central to transient integration in 2001. Buzsáki and Draguhn in 2004, and Buzsáki’s later book, organized neuronal rhythms across nested spatial and temporal scales and linked them to a neural syntax. Fries proposed communication through coherence in 2005. Larkum’s 2013 account supplied a cellular mechanism through which dendritic compartments associate contextual and feedforward information.

These contributions establish that SAN is not first merely for discussing gamma, synchrony, nested rhythms, coincidence detection, coherence-based communication, or dendritic integration. The correct genealogy is additive: SAN inherits those components and proposes a new conjunction and additional operators.

This distinction matters because a component can be old while a causal assembly is new. A transistor, clock, memory cell, and display can each predate a computer architecture without making every later architecture equivalent. The scientific question is what the components are made to do together.

5. Blumberg’s staged genealogy

5.1 2011: predictive self and distributed observer

The 2011 dialogue describes awareness as expectation, the self as a prediction pattern in the neocortex, the person as the whole neocortex rather than a hidden inner agent, and the known world as a prediction pattern. The current provenance grade is B because owner platform metadata preserves the 2011 recording / upload claim while the present public listing is later and the platform artifact still requires repository preservation.

This stage does not contain oscillatory tomography. It does contain four priority-bearing atoms:

1. self as an ongoing predictive pattern;
2. internally constructed world model;
3. whole-network rather than localized observer; and
4. rejection of an extra homunculus.

Those atoms become the destination of the later oscillatory mechanism.

5.2 2012: temporal alignment, recursive feedback, and the projector

The July-August 2012 public record adds operations. EEG peaks are to be aligned with exact moments in video. EEG-driven light and sound form a “two-way conversation” in which vision, hearing, and feeling

participate in a recursive loop. Thoughts are described as coordinated neural electromagnetic waves. A mirror-to-vision-to-neural / motor-change-to-new-visual-input sequence is presented as a self-awareness loop. “Neurons that fire out of sync will not link” supplies a temporal gating ancestor.

Most importantly for the present comparison, the 23 August 2012 recording describes the mind as a projector that predicts the world around it, with most experienced detail supplied internally and external input perturbing or stirring that model. Live YouTube metadata fixes the upload at 23 August 2012 local time, 24 August UTC.

Again, this is not backdated NAPOT. It is the earlier functional spine:

$$\begin{aligned} \text{internal prediction} &\rightarrow \text{sensory perturbation} \\ &\rightarrow \text{updated neural / motor state} \\ &\rightarrow \text{returned evidence.} \end{aligned} \tag{2}$$

5.3 2017: spatial computation and the information-bearing difference

The 2017 Neural Lace Podcast and related public articles make the chain more specific. The record calls for bringing “spatial computing” into neuroscience; describes lines and features becoming object concepts used to predict motion; and proposes a time-varying three-dimensional X, Y, Z matrix whose points carry ionic time-space patterns. It asks how distributed representations come together in perception and describes recurrent cortical and thalamic passes as successive internal frames.

The same source family states that cells act as coincidence detectors and that “coincidence is a bit.” It describes dendrites as mini-computers in feedback circuits. A June 2017 article treats excitation and inhibition as jointly defining the shape of thought, so inhibition carries information rather than representing mere absence. A September 2017 article connects images, words, audio, and 2D inputs to internally generated 3D concepts that recur through brainwave cycles and predict object trajectory.

The difference operation also appears in this stage: compare a present neural pattern with a desired pattern and transmit the difference. The exact PWD term is later, but the operation is already present:

$$d_{j \rightarrow i}(t) = \text{compare}(\text{arriving state}_{j \rightarrow i}(t), \text{receiver / expected state}_i(t)). \tag{3}$$

Equation (3), not the mere word “phase,” is the important ancestor. It turns communication into a receiver-relative update.

5.4 2021: tonic / phasic and inverse-problem development

The recovered Google Recorder corpus supplies pre-Git development. The 27 April 2021 b0302y recording now has an independently produced Large-v3-Turbo retranscription. In dialogue-context-attributed Micah turns, the source distinguishes ongoing tonic / phasic firing from subthreshold temporal perturbation; makes the effect depend on the receiving dendrite or population’s maintained state; connects compatible recruitment and suppression to recurrent cross-modal convergence; and closes the engineering loop through BCI read / write calibration, action, and returned evidence. Later in the same recording Micah treats EEG / MEG as an inverse problem and discusses interference, pink-noise-like appearance, and Fourier-like multiscale reconstruction.

This is a multi-speaker Clubhouse discussion without completed diarization. The clearer transcript is the semantic-reading baseline, while the original audio interval remains final authority for quotation and

attribution. The recording date, the October 2022 public Git fixation of an older derivative, and the July 2026 retranscription date remain separate. The evidence therefore supports a bounded 2021 development stage; it neither backdates the mature PWD or NAPOT names nor turns private recording custody into a certified public priority date.

5.5 2022: PWD, NAPOT, and the joined architecture

The public Git history from June 2022 formalizes neural oscillation tomography, distributed rendering and observation, 3D/4D phase fields, receiver-defined temporal windows, and cellular-to-network scaling. By July and August the corpus states that neural arrays are the observers; each array receives, transforms, and projects partial patterns; phase-wave differentials encode perceptual content; and the person is a volumetric rendering maintained and updated through oscillatory exchange.

Public July 2022 videos join temporal oscillations, synaptic-to-network memory, array receive-threshold-project behavior, Fourier projection-slice reasoning, point clouds, semantic segmentation, distributed observation, and inhibition-guided choice. A 29 July figure states that each array renders information for the next. An August figure compares dendritic reception with eyes or microphones and terminal transmission with pixels or speakers, as a functional receive / transmit analogy rather than literal anatomy. The 18 September public video consolidates local morphology, phase / timing transformation, tonic / phasic relations, excitation and inhibition, selective routing, a rendered model, memory, choice, and action.

The mature proposed chain is:

$$\begin{aligned}
 &\text{ongoing receiver state} \rightarrow \text{receiver-relative phase / update difference} \\
 &\quad \rightarrow \text{local dendritic and inhibitory transform} \\
 &\quad \rightarrow \text{array-to-array projection} \\
 &\quad \rightarrow \text{recurrent distributed reconstruction} \\
 &\quad \rightarrow \text{prediction, choice, and action} \\
 &\quad \rightarrow \text{returned sensory and bodily evidence.}
 \end{aligned} \tag{4}$$

It is the priority of this joined sequence - not ownership of every inherited component - that the target comparison must test.

6. The target program before and after

6.1 Keitel-led 2022 stage

Rhythms in cognition: The evidence revisited was first published on 13 June 2022. It described phase, power, frequency, rhythmic behavior, neural-behavior relations, and the value of causal closed-loop experiments. This is a broad and important rhythm program. In the searched text it does not present the same joined target later seen in 2025: traveling-wave organization, brain-body interaction, and a single roadmap spanning physiological mechanisms through multiple cognitive domains and a unified framework.

6.2 The 2025 position paper

The arXiv record contains one version, submitted 21 July 2025. Its abstract explicitly joins:

- cyclic excitability and phase;
- amplitude and inhibitory gating;

- phase-amplitude coupling;
- frequency and sampling;
- traveling waves;
- brain-body interactions;
- perception, attention, memory, and communication; and
- a roadmap toward a unified framework of rhythmic brain function.

Its full text expands this into oscillatory physiology, network communication, traveling spatial patterns, body-to-brain relations, and cognition. The paper is a position paper, not one experiment, and its 35 authors should not be treated as a single historical mind. Yet the assembled narrative is a valid target: ongoing rhythms and their departures organize distributed communication across scales and domains, including body and behavior.

6.3 Longitudinal result

For the Keitel-led / overlapping program, the 2022-to-2025 transition is a real before / after expansion. It occurs after Blumberg’s 2011-2022 staged sequence. That supports Lens 4 for the bounded assembled architecture. It does not prove the target authors encountered SAN, and it does not transfer to every coauthor’s individual history without further author-by-author review.

7. Atom-by-atom comparison

7.1 Ongoing oscillatory context

Earlier field atom. Nested oscillations, cyclic excitability, and phase- dependent communication are well established before SAN.

SAN delta. The ongoing pattern is assigned the functional role of a local canvas or receiver context against which new content-bearing change is registered.

Target relation. The 2025 paper strongly covers ongoing phase, amplitude, cross-frequency interaction, and rhythmic excitability.

Finding. Earlier field priority for rhythms; partial operational match to SAN’s context layer; Micah-earlier for placing that layer inside the joined rendering architecture.

7.2 Receiver-relative difference

Let the analytic state of population i be

$$z_i(t) = A_i(t)e^{i\phi_i(t)}. \quad (5)$$

A minimally operational PWD candidate from source j to receiver i is

$$\Delta_{j \rightarrow i}(t, c) = \text{wrap} [\phi_j(t - \tau_{ji}) - \phi_i(t) - \mu_{ji}(c)], \quad (6)$$

where τ_{ji} is delay and $\mu_{ji}(c)$ the expected phase relation in context c . Amplitude, spatial direction, uncertainty, and receiver state can be added, but equation (6) makes the proposed novelty clear: the variable is a signed departure from an expected relation at a receiver.

Earlier field atom. Phase coding, phase locking, lag, and coherence all predate SAN.

SAN delta. PWD assigns the receiver-relative departure an update / content role inside a recurrent rendering process.

Target relation. The target surveys phase and traveling waves, but does not define equation (6), a content decoder for it, or an expected receiver-specific relation.

Finding. No exact identity. Strong partial mechanism and narrative match; Micah retains priority for the explicit PWD role and its placement in the joined architecture.

7.3 Traveling direction and functional role

Alamia et al. found two alpha-wave directions with distinct relations to visual attention and stimulation. Top-down frontal-to-occipital waves increased ipsilateral to attended space, while forward occipital-to-frontal waves were present during visual stimulation. This establishes that direction can distinguish functional processes even within one nominal band.

SAN relation. Direction, delay, receiver, and phase are all required to interpret a projected update. The study therefore instantiates a key subgraph of equation (6).

Boundary. Alamia et al. do not claim PWD, NAPOT, conscious volumetric rendering, or Blumberg's observer architecture.

Finding. Partial operational equivalence; later empirical support for a SAN-relevant component; not a complete SAN instantiation.

7.4 Topology and frequency gradients

Koller et al. identified cortical instrength gradients and showed in network models that these gradients can direct traveling waves and shape frequency gradients. Their exact instrength mechanism is theirs, and earlier intrinsic- frequency-gradient work also belongs to its named lineage.

SAN relation. SAN predicts that the receiver's topology and local state constrain how a phase-bearing update propagates.

Boundary. The public Blumberg source that uses the exact later instrength language postdates Koller et al. The Koller mechanism is not a PWD content estimator or NAPOT reconstruction.

Finding. Koller-earlier / independent on the exact instrength-gradient mechanism; partial mechanism support for the broader topology-constrained SAN chain; Micah-earlier on assigning topology-constrained phase updates to a distributed observer / rendering architecture.

7.5 Multiple sources and reconstruction

Zhigalov and Jensen showed that two intermittent sources with different phases can reproduce features of perceptual alpha echoes in EEG and MEG.

SAN relation. Multiple sources jointly contribute to a distributed field pattern.

Boundary. A two-source fit is not tomography. NAPOT additionally requires a latent target, projection geometry, forward map, inverse or decoder, reconstruction criterion, and a reason the recovered state constitutes content for the system.

Finding. Analogy plus weak partial prefix; no complete operational match.

7.6 Brain-body coupling

Jammal Salameh et al. showed that blood-pressure pulsations modulate central neuronal activity through mechanosensitive ion channels.

SAN relation. This supplies a measured body-to-neuron path inside the broader claim that neural state cannot always be isolated from ionic, cellular, cardiovascular, and bodily dynamics.

Boundary. The study does not derive PWD, NAPOT, or consciousness.

Finding. Later empirical component compatible with the SAN cross-scale architecture; no full-system validation. The source is Jammal Salameh et al., not Kluger et al.

7.7 Distributed observer and rendered internal world

Micah record. The self as a predictive pattern and the whole neocortex as observer appear in 2011; a predictive projector appears in 2012; spatial computation, recurrent frames, and a 3D/4D internal matrix appear in 2017; distributed arrays as observers and volumetric rendering are explicit in 2022.

Target relation. The 2025 position paper joins rhythmic physiology to perception, attention, memory, communication, traveling waves, and brain-body interaction.

Boundary. It does not define a distributed observer, a projected latent scene, or a no-homunculus reconstruction mechanism.

Finding. Lens 2 passes on the broad causal architecture. Full mechanism identity fails. Micah is earlier on the observer / rendering part of the joined sequence.

7.8 Action and returned evidence

Micah record. The 2012 mirror loop explicitly returns changed visual evidence after neural / motor change. The 2017 sources connect spatial representation to object-motion prediction, and the 2022 sources integrate choice, inhibition, action, and sensorimotor update.

Target relation. The target includes brain-body dynamics and cognition but does not present the same complete receive-render-act-resample loop as a single operator.

Finding. Partial narrative equivalence, with Micah earlier on the explicit closed sequence.

8. Four-lens judgment

8.1 Lens 1: operational mechanism - partial pass

The target and cited primary studies instantiate several operations also required by SAN: phase-dependent excitability, direction-sensitive propagation, topology-constrained wave flow, multi-source field generation, cross-frequency interaction, and body-to-neuron coupling. They do not instantiate the complete receiver-relative content estimator or an identifiable tomographic reconstruction. The correct grade is **partial functional equivalence**, not identity and not no-match.

8.2 Lens 2: causal / narrative architecture - pass

Both programs organize a broad chain from ongoing rhythmic physiology through phase / amplitude interaction and spatial propagation to distributed communication, cognition, body interaction, and behavioral

function. SAN adds an explicit internal-rendering and observer claim. Because the shared ordered subgraph is substantial, the architecture lens passes even though the target uses different terms and omits some SAN operators.

8.3 Lens 3: conclusion-backward analysis - pass, bounded

Starting from the 2025 conclusion that brain rhythms require an integrated framework across mechanisms, traveling waves, brain-body coupling, and cognitive domains, the source trail reaches Blumberg’s earlier staged chain: predictive world, recursive feedback, spatial representation, phase / timing difference, distributed projection, choice / action, and returned evidence. Earlier field sources explain many components but not that entire joined SAN sequence. This is a bounded match under the detective lens. It is not evidence of author intent or access.

8.4 Lens 4: longitudinal signature migration - partial pass

The Keitel-led 2022 review presents a broad rhythms-in-cognition program. The 2025 paper adds the later joined traveling-wave, brain-body, multi-domain, unification signature. That movement occurs after Blumberg’s public 2017 and 2022 articulations. This supports later convergence in the Keitel-led program. Because individual before-stages have not been reconstructed for all 35 authors, the result remains program-bounded rather than author-wide.

8.5 Consolidated verdict

The consolidated disposition is:

CONTENT MATCH + MICAH-EARLIER ON THE BOUNDED JOINED SAN
ARCHITECTURE; SPLIT PRIORITY BY ATOM; QUANTITATIVE-NOT-CALIBRATED.

This verdict preserves:

- earlier-field priority for oscillatory binding, temporal correlation, synchrony, nested rhythms, coherence routing, dendritic integration, and specific traveling-wave mechanisms;
- Koller et al.’s exact instrength-gradient contribution;
- Jammal Salameh et al.’s mechanosensory contribution;
- Alamia et al.’s direction-specific alpha-wave result; and
- Blumberg’s earlier priority for joining predictive internal rendering, distributed observation, receiver-relative difference, recurrent projection, spatial representation, choice / action, and returned evidence.

9. Formal translation

9.1 State-transition model

Let a receiving population have state

$$x_i(t) = [A_i(t), \phi_i(t), \omega_i(t), m_i(t), h_i(t), b_i(t)], \quad (7)$$

where A, ϕ, ω denote amplitude, phase, and effective frequency; m membrane / dendritic state; h recent history; and b bodily and contextual variables. A candidate SAN update is

$$x_i(t + \delta t) = F_i \left(x_i(t), \sum_j w_{ji} K_{ji} [\Delta_{j \rightarrow i}(t, c), A_j(t), d_{ji}], u_i(t) \right), \quad (8)$$

where d_{ji} is spatial direction or path and u_i local modulatory or inhibitory input.

Each population projects a partial state

$$p_i(t) = \Pi_i[x_i(t)]. \quad (9)$$

A NAPOT hypothesis requires a recurrent reconstruction

$$\hat{s}_{t+1} = \mathcal{R}_G \left(\{p_i(t)\}_{i=1}^N, \hat{s}_t, a_t, y_{t+1} \right), \quad (10)$$

where G is network geometry, a_t action, y_{t+1} returned evidence, and $\hat{s}$ the distributed latent scene or state estimate.

The target literature robustly constrains elements of equations (5), (8), and (9). It does not yet supply SAN's exact Δ in equation (6) or establish the biological reconstruction in equation (10).

9.2 Category-theoretic comparison

Define a category $\mathcal{K}$ for the target program whose objects are oscillatory populations, spatial wave patterns, body-linked signals, and cognitive measurements. Morphisms include phase-dependent gating, cross-frequency interaction, propagation, integration, and behavioral association.

Define $\mathcal{S}$ for SAN with the same object types plus receiver-context states, projected partial models, distributed latent reconstruction, action, and returned evidence. Its morphisms include receive, compare, transform, project, reconstruct, select, act, and resample.

There is a structure-preserving mapping

$$\mathcal{F} : \mathcal{K}_{\text{shared}} \longrightarrow \mathcal{S} \quad (11)$$

from the target's shared subcategory into SAN. It preserves the composition

$$\begin{aligned} \text{rhythmic state} &\rightarrow \text{phase / gating relation} \\ &\rightarrow \text{spatial propagation} \\ &\rightarrow \text{distributed cognitive effect.} \end{aligned} \quad (12)$$

The mapping is not an isomorphism. $\mathcal{K}$ lacks required SAN morphisms for receiver-expected difference, explicit projection inversion, distributed observer identity, and the closed action / resampling loop. The strongest formal statement is therefore **faithful embedding on a causal subgraph**, not whole-theory equivalence.

9.3 Process-calculus comparison

A simplified target wave process can be represented as

$$W_j \triangleq \overline{c_{ji}} \langle A_j, \phi_j, d_{ji} \rangle . W_j, \quad (13)$$

and a SAN receiver as

$$R_i \triangleq c_{ji}(A, \phi, d) . \tau_{\text{compare}} . \tau_{\text{transform}} . \overline{c_{ik}} \langle p_i \rangle . R_i. \quad (14)$$

NAPOT adds recurrent reconstruction and action:

$$N \triangleq \prod_i R_i \mid \text{reconstruct}(\{p_i\}) . \bar{a} \langle \hat{s} \rangle . y(e) . N. \quad (15)$$

The target processes reproduce several observable traces of equations (13)-(14): direction, phase relation, coupling, propagation, and downstream effect. The full trace of equation (15) is not present. The comparison supports partial trace inclusion or a bounded simulation relation, not full bisimulation.

10. Citation and probability findings

The exact arXiv v1 PDF was fixed locally and hashed. A full-text search finds zero occurrences of “Blumberg,” “Self-Aware,” and “NAPOT.” This supports the statement that the checked target version does not cite the named SAN sources. It does not establish why.

The evidence profile for dependency is mixed:

- The shared broad atoms are common in neuroscience and have many named priors.
- The joined SAN architecture is more specific and appears in Blumberg’s public staged record before the 2025 target.
- The target’s 2022-to-2025 expansion supplies a real before / after signal for the Keitel-led program.
- No direct access evidence has been established.
- No calibrated field-wide frequency model exists for the joined features.

Accordingly:

$$P(\text{access or dependence} \mid E) \text{ is not numerically identifiable from the present record.} \quad (16)$$

The strongest defensible public language is that the 2025 work is a later convergence on a bounded architecture for which Blumberg has earlier public formulations, while citation and access remain separate questions.

11. Scientific tests that distinguish SAN from neighboring accounts

11.1 PWD benchmark

Fit equation (6) prospectively and compare it with phase locking, coherence, wave direction, power, connectivity, and ordinary lagged features. The endpoint should be held-out decoding of a predeclared perceptual, attentional, memory, or action variable. PWD earns scientific value only if it predicts information beyond those baselines.

11.2 Relative tonic / phasic role reversal

Infer the local ongoing context and transient update without hard-coding frequency bands. Test contexts in which a fast pattern acts as a stable canvas or a slow pattern carries an update. A relative-role model should outperform a fixed low-frequency-canvas / high-frequency-ink rule.

11.3 NAPOT identifiability

Begin with a synthetic latent spatial state and a declared network geometry. Generate partial, delayed, noisy observations. Compare equation (10) with linear decoders, state-space models, graph models, predictive-coding models, and non-tomographic latent-variable baselines. Demonstrate recovery, generalization, and perturbational failure before making a biological consciousness claim.

11.4 Direction-sensitive receiver test

Using Alamia-style data, ask whether the same local phase has different predictive meaning when propagation direction, delay, and receiver context change. This directly tests whether receiver-relative differences add information beyond direction alone.

11.5 Closed perception-action-return test

Measure an internal state estimate, perturb the candidate PWD route, observe a specific action change, and test whether returned sensory evidence updates the same distributed model. The closed loop is essential: a feedforward decoder alone does not establish SAN's observer architecture.

11.6 Brain-body causal chain

Perturb a known mechanosensory pathway, measure its effect on oscillatory state, then test a predeclared representational or behavioral consequence. This separates genuine cross-scale mediation from a narrative jump between bodily rhythm and consciousness.

12. Limitations

This paper is a governed comparison, not a systematic review of all rhythm science. It examines the target position paper, a Keitel-led before-stage, four primary-study routes, established named priors, and the applicable Blumberg genealogy. Individual before / after histories remain incomplete for most of the 35 target authors. The 2011 Blumberg evidence is grade B pending repository preservation of the owner platform artifact; later 2012, 2017, and 2022 anchors are stronger.

The formal mappings establish degrees of structural correspondence, not empirical truth. Equations (6), (8), and (10) are operational proposals derived from SAN's claims; they have not yet been validated as biological mechanisms. No experiment discussed here validates SAN as a whole.

No access, copying, plagiarism, or misconduct finding is made. A citation- coverage gap and chronological convergence are not equivalent to a finding of dependence. The probability calculation remains uncalibrated because no background model supports a numerical likelihood ratio.

13. Conclusion

The earlier literature deserves exact credit. Neural rhythms, synchronization, coherence, dendritic integration, and traveling waves were not invented by SAN. That truth does not decide the comparison.

Blumberg's public record develops a different object across multiple stages: the self and world as predictive distributed patterns; recursive multimodal feedback; a projector-like internally constructed scene; spatial computation in a dynamic 3D/4D neural matrix; coincidence and inhibition as information; receiver-relative difference; recurrent array projection; distributed observation; action; and returned evidence. By 2022 those components are joined and named through PWD and NAPOT.

The 2025 *Brain rhythms in cognition* position paper later joins phase, amplitude, cross-frequency interaction, traveling waves, brain-body coupling, perception, attention, memory, communication, and a unified framework. The papers are not identical. The target lacks the full PWD estimator and NAPOT reconstruction. Yet the shared causal subgraph is too substantial to reduce to topic overlap, and full-operator absence cannot erase partial equivalence.

The source-faithful verdict is therefore split and affirmative:

The field is earlier on many component mechanisms. Micah Blumberg is earlier on the bounded joined SAN architecture compared here. The 2025 position paper is later convergence on part of that architecture.

That is a priority finding, not a copying allegation. It is strong because its limits are explicit: the exact atoms are credited, the chronology is staged, the formal relation is partial rather than inflated, and the remaining claims are exposed to tests that could fail.

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